

TELECOM CHURN PREDICTION

What I built, what the results mean, and what I would improve next

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Status	Submission-ready

Project overview

What I set out to do: I wanted to build a simple tool that could help a telecom retention team identify customers who may need attention. I explored the data, trained a churn model, checked its performance on unseen records and built a Streamlit app for individual predictions.

Current status: The project is ready for submission. The training process is separate from the app, the model is saved with its preprocessing steps, and the final results come from a holdout test set.

Validated model performance

Metric	Original 50% threshold	Revised 33% threshold
Accuracy	80.55%	76.30%
Precision	65.72%	54.00%
Recall	55.88%	72.19%
F1 score	60.40%	61.78%
ROC-AUC	84.19%	84.19%

What this means: changing the threshold does not change how the model ranks customers, so ROC-AUC remains the same. At the 33% review threshold, the model found 270 of the 374 churners in the test

set. At the usual 50% threshold, it found 209. The benefit is better recall; the cost is contacting more people who would have stayed.

Actual / predicted	Stayed	Churned
Actual stayed	805	230
Actual churned	104	270

How I improved the project

- I removed the exposed ngrok token and the Colab-only tunnel cells from the notebook.
- I reorganised the notebook so another person can follow it from data checks through to the final evaluation.
- I created train_model.py and saved the complete pipeline, so the app uses the same preparation steps as the model evaluation.
- I simplified the Streamlit app into three areas: an overview, a customer prediction form and a model-performance page.
- I added clear setup instructions, automated checks, security guidance and a submission checklist.
- I made the limits of the public dataset clear instead of presenting it as Nigerian operator data.

Data and modelling quality

Check	Result	Assessment
Rows / fields	7,043 / 21	Adequate for a capstone demonstration
Duplicate rows	0	Pass
Missing values	11 TotalCharges after numeric conversion	Handled inside the pipeline
Class balance	26.54% churn	Use recall, F1, ROC-AUC and average precision
Data leakage	Separate train/validation/test sets	Pass
Deployment parity	Saved full preprocessing + classifier pipeline	Pass

What I noticed in the data

- Customers on month-to-month contracts had the highest churn rate in the sample.
- Newer customers appeared more often among churners, which suggests that the early customer experience deserves attention.
- Higher monthly charges, electronic-check payments, fibre-optic service and missing support/security services were useful clues for the model.
- These are patterns, not proven causes. They are a starting point for customer conversations and further testing.

Risks and limitations

Risk	Why it matters	Required control
External sample data	May not represent Nigerian customers or current market conditions	Retrain on recent, consented local data
Threshold economics	33% creates more false positives	Set threshold using value, offer cost, and team capacity
Fairness and privacy	Risk scores can lead to unequal treatment or excessive data use	Segment testing, access controls, retention policy, human review
Model drift	Customer behaviour and offers change over time	Monitor calibration, recall, and churn rate after launch
Causal overreach	Predictors do not prove why churn occurs	Validate retention actions with controlled tests

Submission benchmark

Area	Rating	Evidence
Problem framing	Strong	Clear churn objective and retention use case
Data preparation	Strong	Typed conversion, pipeline imputation, ID exclusion
EDA	Strong	Decision-relevant segment and

		tenure analysis
Model evaluation	Strong	Holdout metrics, confusion matrix, threshold trade-off
Application	Strong	Saved model, structured interface, actionable output
Documentation	Strong	Reproducible setup, scope, limitations, checklist
Security	Improved	Credential removed; secrets guidance added
Identity metadata	Complete	Name, Fellow ID, email, and gender updated consistently

Final actions before submission

1. Upload the cleaned ZIP to the repository; do not restore the old notebook containing the token.
2. Run `python train_model.py` and `streamlit run app.py` in the target environment.
3. Add the final GitHub repository link and deployed Streamlit link where requested.
4. Rotate or revoke the exposed ngrok token if it is still active.

My conclusion

This project taught me that a useful machine-learning solution is more than a high accuracy score. The data preparation must be repeatable, the evaluation must match the business problem, and the result must be explained honestly. The model is not ready for a real telecom company without local data and further testing, but it provides a complete working example of how churn prediction can support a retention team.